SERVOS + GPIO

USB HUB

DEBUG HEADER
BOOT
RESET
STATUS LED

ESP32-S3

UART TO PI

ULTRASONIC
I2C
UART

SENSORS

POWER

SCREW TERMINAL INPUT

5V USB RAIL

SERVO POWER RAIL

3.3V LOGIC

FUSE / PROTECTION

UPSTREAM TO PI

USB-A 1
USB-A 2
USB-A 3
USB-A 4

USB-C 1
USB-C 2
USB-C 3

ID_SD and ID_SC PINS:
These pins are reserved for HAT ID EEPROM.

At boot time this I2C interface will be interrogated to look for an EEPROM that identifes the attached board and allows automagic setup of the GPIOs (and optionally, Linux drivers).

Bridging JP1 enables write protection.

DO NOT USE these pins for anything other than attaching an I2C ID EEPROM. Leave unconnected if ID EEPROM not required.

Additional pin functions:
https://pinout.xyz/

External3V3 input: panel buck -> J10 pin2 BOARD_3V3; J10 pin1 GND. Original AP2112 is REMOVED, not paralleled.
UART is noninverting. GPIO39 high means USB VBUS present. Pi +3V3 remains separate from BOARD_3V3.

Sheet: /	
File: test1.kicad_sch	
Title: Modular Pi HAT – rev23 external3V3 / protected UART and VBUS	
Size: A3	Date: 15 nov 2012
KiCad E.D.A. 10.0.7	Id: 1/1

Rev: 23 engineering prototype